Why higher $\alpha\%$ confidence interval is not necessarily better?

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(and 𝕏)

Abstract. Recently, someone had asked me "if we want to estimate some parameter θ, say $\hat{\theta}$, we compute $\alpha\%$ confidence interval to approximate the interval at which the exact value for the parameter could lie in. So, doesn't higher $\alpha\%$ provide better approximation, if so why don't researchers always use 99% confidence interval, instead of 95% or 97.5% which we see often?" Personally, I thought it was a valid and really good question which is extremely counter-intuitive and non-trivial to think about. So I decided to write a short note on it.

For learning Statistical theory (or mathematical statistics) with high rigour, I have mentioned a extensive (but obviously not exhaustive) list of resources (mostly in order of difficulty, i.e., easy to difficult). According to me the best book to start from is "Casella and Berger" [5], but might be hard!

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1. Introduction

The intuition of the student who asked me is *partially* correct: a 99% confidence interval has a higher probability of covering the true parameter than a 95% confidence interval. However, it generally provides a less precise estimate of that parameter. What do I mean by that?

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This is one of the fundamental trade-offs in statistical theory:

Confidence versus Precision

I will first try to explain this intuitively and then mathematically.

2. Why isn't a 99% CI always better?

Suppose we want to estimate an unknown population parameter θ, and we obtain an estimator $\hat{\theta}$. Usually, we construct a confidence interval of the following form:

$$CI_{1-\alpha} = [L(X), U(X)],$$

where X denotes the observed data.

There are two nice properties of this interval:

(1) *High coverage probability:* We want the confidence interval to contain the true parameter θ frequently.
(2) *Small interval width:* We want the interval to be as narrow as possible, so that our estimate is informative.

Unfortunately, these objectives generally conflict. Increasing the confidence level usually requires widening the interval. This is sad and very counter-intuitive!! :(:(

3. A numerical example

Suppose we estimate a parameter and obtain

$$\hat{\theta} = 10 \qquad \mathrm{SE}(\hat{\theta}) = 2$$

Assume the estimator is normally distributed and its standard error is known. Now, we will construct a two-sided confidence intervals (I hope you know how to do that!) and compare to the fixed value for $\hat{\theta}$:

Confidence level	Confidence interval	Total width
90%	$[6.71, 13.29]$	6.58
95%	$[6.08, 13.92]$	7.84
97.5%	$[5.52, 14.48]$	8.96
99%	$[4.85, 15.15]$	10.30

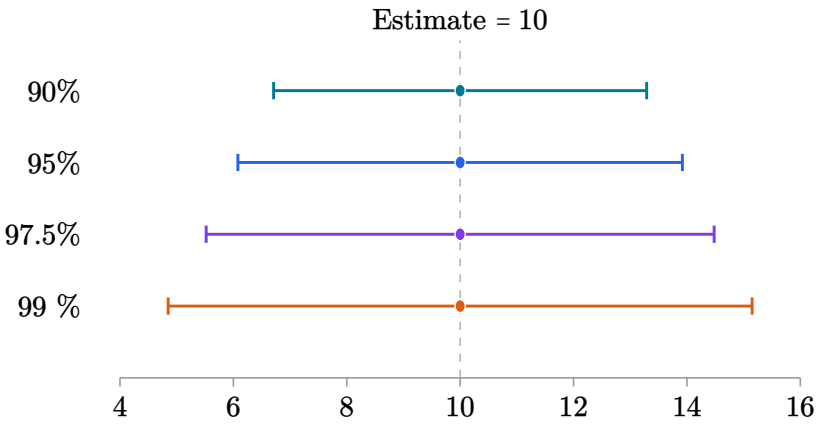
Table 1. Confidence intervals and their widths at different confidence levels.

Notice something important? The 99% confidence interval covers a much larger range of potential parameter values than the 95% interval.

Consequently:

- The 99% interval has better coverage.
- The 95% interval has better precision.

Therefore, neither interval is universally better than the other.



4. Isn't it counter-intuitive?

But doesn't higher CI provide more narrower width? It is so counter-intuitive. It seems counter-intuitive because we naturally associate higher confidence with greater accuracy. As discussed previously, in statistical inference, confidence and precision are two different concepts.

> A higher confidence level means we want to be more certain that our interval captures the true parameter θ. To achieve that, we must generally make the interval wider, not narrower.

Let's think of a very simple example again. Suppose an unknown parameter θ lies somewhere between 0 and 100, and we want to construct an interval containing it.

Consider two possible intervals:

- Interval A: $[49, 51]$
- Interval B: $[10, 90]$

Which interval is more likely to contain θ, assuming we have no additional information? Clearly, Interval B covers more possible values of θ.

But which interval provides more precise information about θ? Clearly, Interval A, because it restricts θ to a much smaller range.

That is the distinction:

- **Confidence:** How reliably does our procedure capture the true parameter?
- **Precision:** How narrowly does our interval localize the parameter?

A wider interval sacrifices precision to increase coverage.

4.1. **Extreme case?**

Consider the extreme case, i.e., 100% confidence. Imagine that our parameter is any real number:

$$\theta \in \mathbb{R}.$$

We could construct the confidence interval

$$\mathrm{CI}_{100\%} = (-\infty, \infty).$$

This interval would certainly contain θ!

Its coverage probability is

$$\mathbb{P}_\theta(\theta \in \mathbb{R}) = 1.$$

But is this interval useful? *Not at all!* It tells us absolutely nothing about the location of θ.

Maximizing confidence alone is not sufficient. An interval can have perfect coverage while providing essentially no information about the unknown parameter.

4.2. **Visualize it using the normal distribution.** Suppose

$$\hat{\theta} \sim N(\theta, \sigma^2/n).$$

The following illustration shows the central regions under the standard normal distribution.

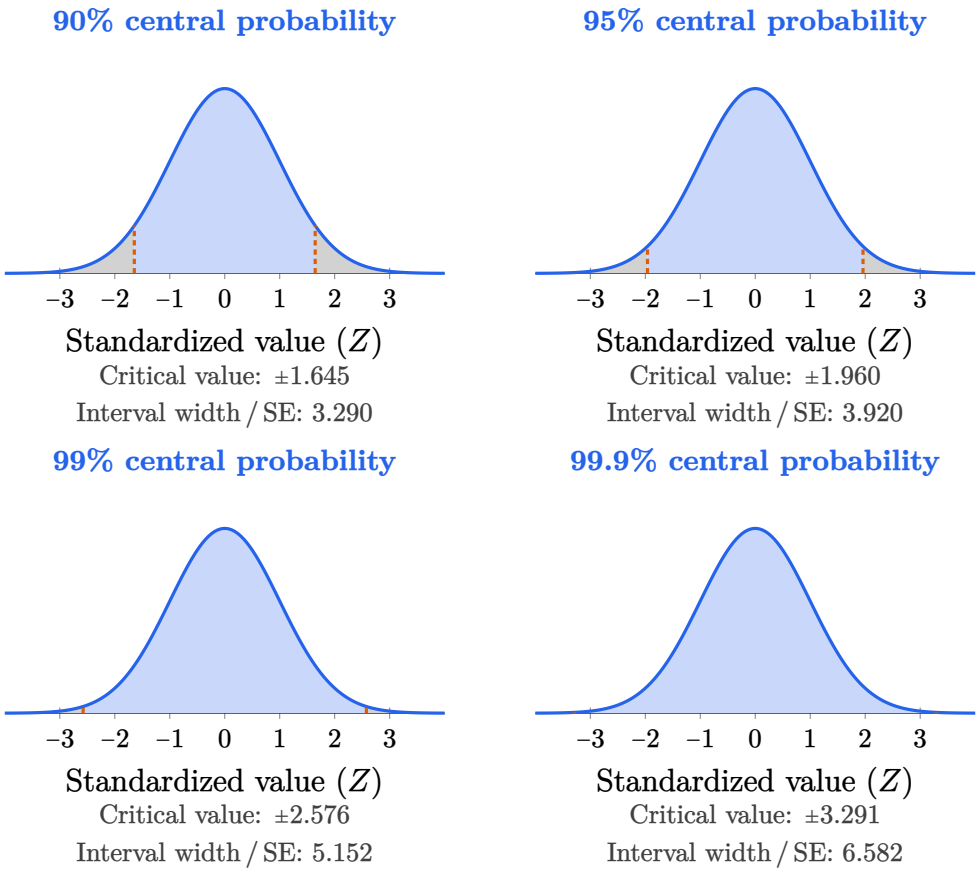


FIGURE 1. Standard normal density with the central probability shaded for confidence levels of 90%, 95%, 99%, and 99.9%.

As you increase the confidence level from 90% to 99.9%, notice that the shaded region expands outward.

The mathematical reason is straightforward:

$$\mathrm{CI}_{1-\alpha} = \left[\hat{\theta} - z_{1-\alpha/2}SE, \quad \hat{\theta} + z_{1-\alpha/2}SE\right].$$

For example,

$$z_{0.975} = 1.96, \qquad z_{0.995} = 2.576.$$

Consequently, the 99% confidence interval must extend further from $\hat{\theta}$ than the 95% confidence interval.

4.3. **The most important conceptual distinction.** Consider the following two questions:

- How confident are we that the interval-generating procedure captures the true parameter? Increasing the confidence level helps.
- How precisely does the resulting interval identify the location of the true parameter? Reducing the interval width helps.

These are not equivalent questions.

In fact, for a fixed sample size and a conventional confidence interval procedure, improving one generally comes at the expense of the other.

This is precisely why mathematical statisticians investigate optimal confidence intervals: they seek intervals with the smallest possible expected width subject to a specified minimum coverage probability.

The student's intuition that better statistical inference should produce narrower intervals is correct when better inference means obtaining more information, such as through a larger sample size. But simply demanding a higher confidence level does not give us more information about θ; it makes us more conservative about which parameter values we exclude.

5. Rigorous Reasoning

For convenience, assume that $\hat{\theta}$ is a BLUE (Best Linear Unbiased Estimator) by satisfying the Gauss-Markov conditions. Suppose

$$\hat{\theta} \sim \mathcal{N}\left(\theta, \frac{\sigma^2}{n}\right),$$

where σ is known.

Then

$$Z = \frac{\hat{\theta} - \theta}{\sigma/\sqrt{n}} \sim \mathcal{N}(0,1).$$

For a confidence level $1-\alpha$, we know

$$\mathbb{P}\left(-z_{1-\alpha/2} \le Z \le z_{1-\alpha/2}\right) = 1-\alpha.$$

Rearranging gives

$$\mathbb{P}_\theta\left(\hat{\theta} - z_{1-\alpha/2}\frac{\sigma}{\sqrt{n}} \le \theta \le \hat{\theta} + z_{1-\alpha/2}\frac{\sigma}{\sqrt{n}}\right) = 1-\alpha.$$

Thus, our confidence interval is

$$\boxed{CI_{1-\alpha} = \left[\hat{\theta} - z_{1-\alpha/2}\frac{\sigma}{\sqrt{n}},\ \hat{\theta} + z_{1-\alpha/2}\frac{\sigma}{\sqrt{n}}\right].}$$

Its width is

$$\boxed{W_{1-\alpha} = 2z_{1-\alpha/2}\frac{\sigma}{\sqrt{n}}.}$$

As the confidence level $1-\alpha$ increases, the critical value $z_{1-\alpha/2}$ increases.

Hence,

$$\boxed{\text{Higher confidence} \implies \text{Greater interval width.}}$$

This is why researchers do not automatically select the largest confidence level.

6. Why is 95% so common in statistical research?

To be honest, there is nothing mathematically nice about 95%. It is largely a traditional and disciplinary convention that offers a nice middle ground between coverage and precision. The appropriate confidence level depends on the research objective.

The appropriate confidence level depends on the research objective. For example, if missing the true parameter by mistake causes a huge blunder, a researcher might choose 99% confidence. If obtaining a narrow interval is more important, a 90% or 95% interval might be appropriate.

Since we are discussing about confidence interval, we should also think about hypothesis testing. Isn't it obvious!! Under standard duality between tests and confidence intervals, a two-sided 95% confidence interval corresponds to a two-sided hypothesis test with significance level $\alpha = 0.05$. A 99% interval corresponds to a test with significance level $\alpha = 0.01$. The 99% procedure reduces the probability of a Type I error under the null hypothesis, but, holding sample size and other assumptions fixed, it generally reduces the test's power.

7. Can we increase confidence without sacrificing precision?

Yes, by increasing the sample size. But why? Recall

$$W_{1-\alpha} = 2z_{1-\alpha/2}\frac{\sigma}{\sqrt{n}}.$$

Increasing n decreases the width.

For example, to construct a 99% interval with the same width as a 95% interval, assuming the same population variability, we require

$$2(2.576)\frac{\sigma}{\sqrt{n_{99}}} = 2(1.960)\frac{\sigma}{\sqrt{n_{95}}}.$$

Therefore,

$$\boxed{n_{99} = \left(\frac{2.576}{1.960}\right)^2 n_{95} \approx 1.73\, n_{95}.}$$

In other words, we need approximately *73% more observations* to achieve a 99% interval of the same width as the original 95% interval in this model.

I hope this explains why sample size is so important for statistical inference.

8. One important theoretical distinction

Recall that the student mentioned that "the confidence interval approximates where the exact parameter could lie." There is a subtle issue here that is important in *frequentist* statistical theory.

The parameter θ is fixed but unknown. The confidence interval is random because it depends on the sample X.

Thus, the correct frequentist coverage statement is

$$\boxed{\mathbb{P}_\theta\left(\theta \in \mathrm{CI}_{1-\alpha}(X)\right) = 1 - \alpha.}$$

For a 95% confidence procedure, this means that if we repeatedly sampled data and constructed intervals using the same procedure, 95% of those intervals would contain the true parameter.

It does *not*, under the usual frequentist interpretation, mean that there is a 95% probability that the fixed parameter lies in the particular interval already computed.

That latter type of probability statement is associated with Bayesian credible intervals, where uncertainty about the parameter is represented through a posterior distribution. Don't worry if this idea goes over your head!!

9. The deeper statistical-theory perspective

In mathematical statistics, confidence interval construction is not simply a problem of maximizing coverage probability.

It can instead be formulated as a constrained optimization problem.

For instance, among all interval procedures $C(X)$ satisfying

$$\inf_{\theta \in \Theta} \mathbb{P}_\theta \left(\theta \in C(X)\right) \geq 1 - \alpha,$$

we might seek a procedure minimizing expected interval length:

$$\boxed{\min_{C} \mathbb{E}_\theta \left[\text{Length}(C(X))\right]}$$

subject to the coverage constraint, or minimize a weighted or worst-case version of expected length across values of θ.

This leads to the theory of *optimal confidence sets*, which examines how to construct intervals that are as informative as possible while preserving prescribed coverage guarantees. Different optimality criteria can lead to different procedures.

10. Conclusion

A higher confidence level does not necessarily mean a statistically better estimate. It means stronger coverage guarantees, ordinarily at the cost of a less informative interval. The goal of statistical inference is to balance reliability and precision, rather than maximize the confidence percentage alone.

Declaration on the usage of AI

No large language models (LLMs) such as ChatGPT, Claude, Gemini or any other AIs were used while writing this essay. This only contains by ideas and explanations. So take it with a slight pinch of salt. :) :)

I believe I am pretty experienced to write on this topic, so no worries. But I would like to confess, that LLMs were only used for generating LaTeX for the diagrams, that I intended to make for intuitive visualization.

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